Thin-Film Interference

> **SYNOPSIS**
>
> Soap bubbles and oil slicks show colours no pigment put there. Two rays leave the film — one off the front surface, one off the back — and interfere. But the obvious calculation gives the wrong answer for a very thin film, and fixing it needs something new: **reflection can flip a wave over**.

BEFORE YOU START — PREDICT

A soap bubble gets **thinner and thinner** just before it bursts, so the two rays travel almost exactly the same distance. The path difference goes to zero. So the thinnest part should be **brightest**. In fact it goes **black**. Why?

PERFORMANCE EXPECTATIONS

1. **Identify** the two interfering rays in a thin film and say which surface each reflected from.
2. **Determine** whether a reflection inverts the wave, from the indices on either side of it.
3. **Combine** the path difference $2t$ with the phase changes on reflection to get the correct condition.
4. **Compute** film thicknesses for constructive or destructive interference, using $\lambda_n = \lambda_0/n$.

3. Further topics — thin-film interference

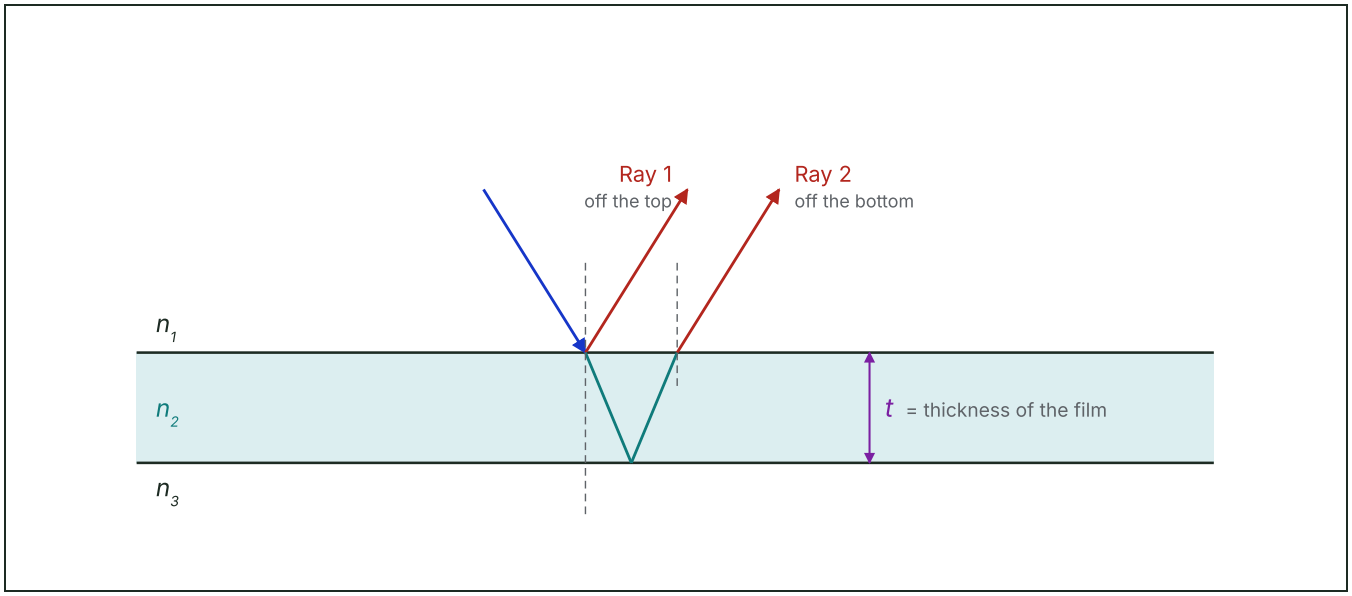


Figure 1. Two rays reach your eye from a thin film: one bounced off the **top** surface, one that went through, bounced off the **bottom** surface and came back out. They are parallel, they overlap, and they have travelled

different distances — so they interfere. **That is the whole mechanism.**

Your job on Figure 2. The rays are drawn for you in both panels. At each of the four reflection points, decide whether the wave **inverts** — compare the index it is in with the index it is bouncing off — and mark it. Then say which row of the table each panel belongs to.

In a **soap bubble**, or any very thin layer, the dark spots are where the **bubble is thinnest** — and that is where it breaks.

THE QUESTION THIS RAISES

Q. If t is very small — a very thin film — the path-length difference between Ray 1 and Ray 2 is small. So shouldn't there be **constructive** interference?

A. Phase change upon reflection!!

Reflection can flip the wave over

Specifically, **reflection causes a 180° phase change**. We observed this clearly in mechanical waves — a pulse on a string reflecting off a fixed end comes back upside down. For a light wave, the deciding parameter is n, the refractive index.

- **Low to high n** → 180°, or π rad, phase change.
- **No phase change** when reflecting from a **lower** index.
- **Refraction causes no phase change at all** — only reflection does this.

And the useful part: a **180° phase change is equivalent to ±λ/2 of distance travelled**, so it can be added straight into the path-length bookkeeping rather than tracked separately.

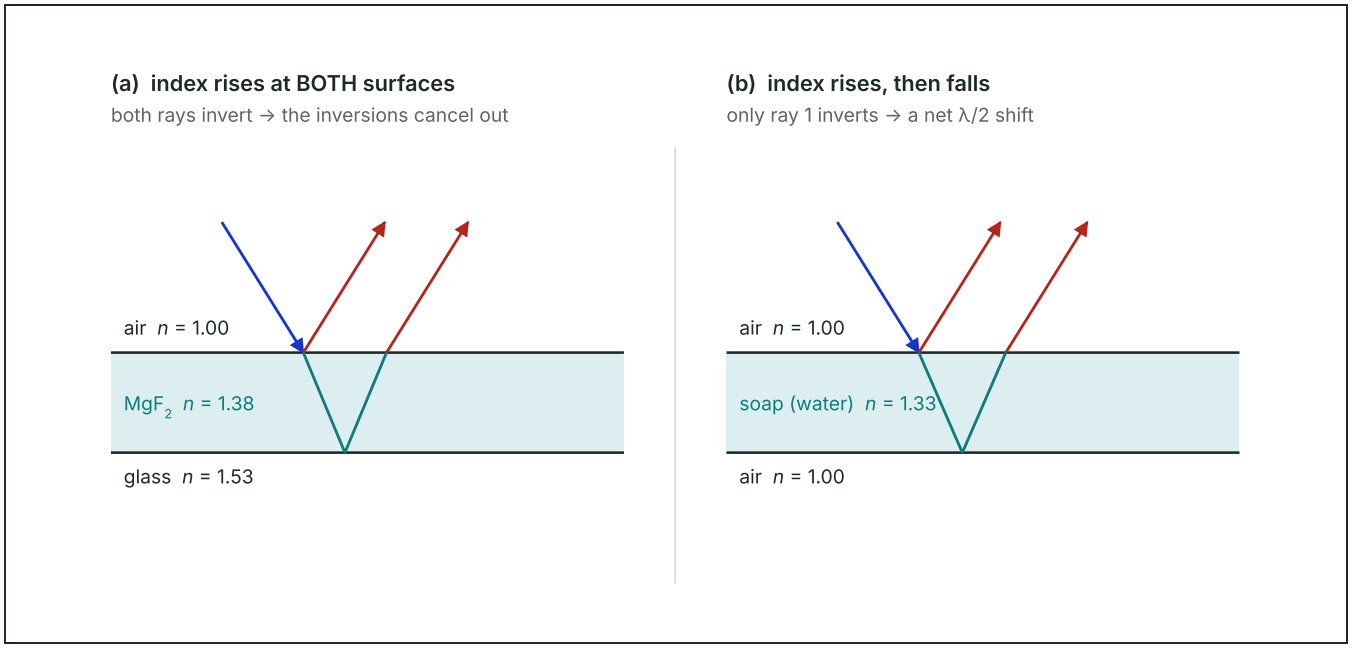


Figure 2. Identical geometry, different materials — and **opposite rules**. Count the inversions before you write anything down: **two inversions or none** and they cancel; **exactly one** and the film behaves as though it were half a wavelength thicker than it is.

Putting the two together

If the thickness t is **not** negligible, you must consider the path-length difference **as well as** the phase change on reflection. Remember that a path-length difference of $m\lambda$ is constructive and $(m + ½)\lambda$ is destructive — and inside the film the wavelength is not the one you started with:

$$\lambda_n = \lambda_0 / n$$

The ray goes down and back, so the extra path is $\mathbf{2}\boldsymbol{t}$. Which condition that satisfies depends on how many inversions happened:

Inversions	Constructive	Destructive
Two, or none they cancel out	$2t = \lambda_n, 2\lambda_n, 3\lambda_n\ldots$	$2t = \lambda_n/2, 3\lambda_n/2, 5\lambda_n/2\ldots$
Exactly one a net λ/2 shift	$2t = \lambda_n/2, 3\lambda_n/2, 5\lambda_n/2\ldots$	$2t = 0, \lambda_n, 2\lambda_n, 3\lambda_n\ldots$

The two rows are each other's opposites. Nothing else in this note is worth memorising if you can count inversions and read the right row.

Worked example 1 — an anti-reflection coating

A layer of $\mathbf{MgF_2}$ (magnesium fluoride, $n = 1.38$) is deposited on glass ($n = 1.53$) in air. **What is the thinnest $\mathbf{MgF_2}$ layer that will limit the reflection of 550 nm light?**

Worked example 2 — a soap bubble

A soap bubble is water ($n = 1.33$) with air on both sides. Find the **three smallest thicknesses** that give **constructive** interference for 650 nm light, and the three smallest that give **destructive** interference.

1. Count the inversions first. This film is *not* the coating above, and the answer changes because of it.

THE PUNCHLINE

Count the inversions, then pick the row

Back to the prediction: the thinnest part of the bubble goes black because **one** reflection inverts and the other does not. With $t \to 0$ the two rays are still half a wavelength out of step, and they cancel. The blackness just before a bubble bursts is **direct evidence of the phase change on reflection**.

Looking ahead

A film gives two paths that differ by microns. **Note 13** gives them two paths that differ by whatever you like — an instrument built to make the path difference adjustable, and accurate to a fraction of a wavelength.

2. A 500 nm wave enters a film of $n = 1.25$. What is its wavelength inside the film, in nm?

nm

3. In a film where **exactly one** of the two reflections inverts, a thickness of $t = 0$ gives:

○ Constructive interference — the paths are identical. ○ Destructive interference — the lone inversion is still a λ/2 shift. ○ Neither; there is no film, so nothing happens. ○ It depends on the wavelength.

4. What is the thinnest anti-reflection coating of $n = 1.25$ on glass ($n = 1.52$) for 600 nm light, in nm?

nm

Check Clear

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